Mini-RAG Test Document

Artificial Intelligence

Artificial Intelligence (AI) is the field of building systems that perform tasks requiring human intelligence such as reasoning, learning, language understanding, and decision making.

Machine Learning

Machine Learning is a subset of AI where models learn patterns from data instead of being explicitly programmed.

Deep Learning

Deep Learning uses multi-layer neural networks. Popular frameworks include PyTorch and TensorFlow.

Embeddings

Embeddings convert text into dense numerical vectors. Similar meanings produce vectors that are close in vector space.

Vector Databases

A vector database stores embeddings and enables efficient similarity search. Chroma is a lightweight vector database commonly used in RAG projects.

Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with a Large Language Model. First, relevant document chunks are retrieved. Then those chunks are provided as context to the LLM, which generates an answer grounded in the retrieved information.

Chunking

Large documents are split into smaller chunks before embedding. A chunk size of around 1000 characters with 200 characters of overlap is a common starting point.

MMR Retrieval

Maximal Marginal Relevance (MMR) balances relevance and diversity. It retrieves a larger candidate set and selects diverse results to reduce duplicate context.

Large Language Models

Examples include Mistral, Gemini, Grok, GPT, and Llama. In a RAG system, the LLM should answer using only the retrieved context.

Sample FAQ

Q: What is RAG?
A: Retrieval-Augmented Generation.

Q: What vector database does this document mention?
A: Chroma.

Q: Which frameworks are mentioned for deep learning?
A: PyTorch and TensorFlow.

Q: What does MMR improve?
A: It improves retrieval diversity while maintaining relevance.